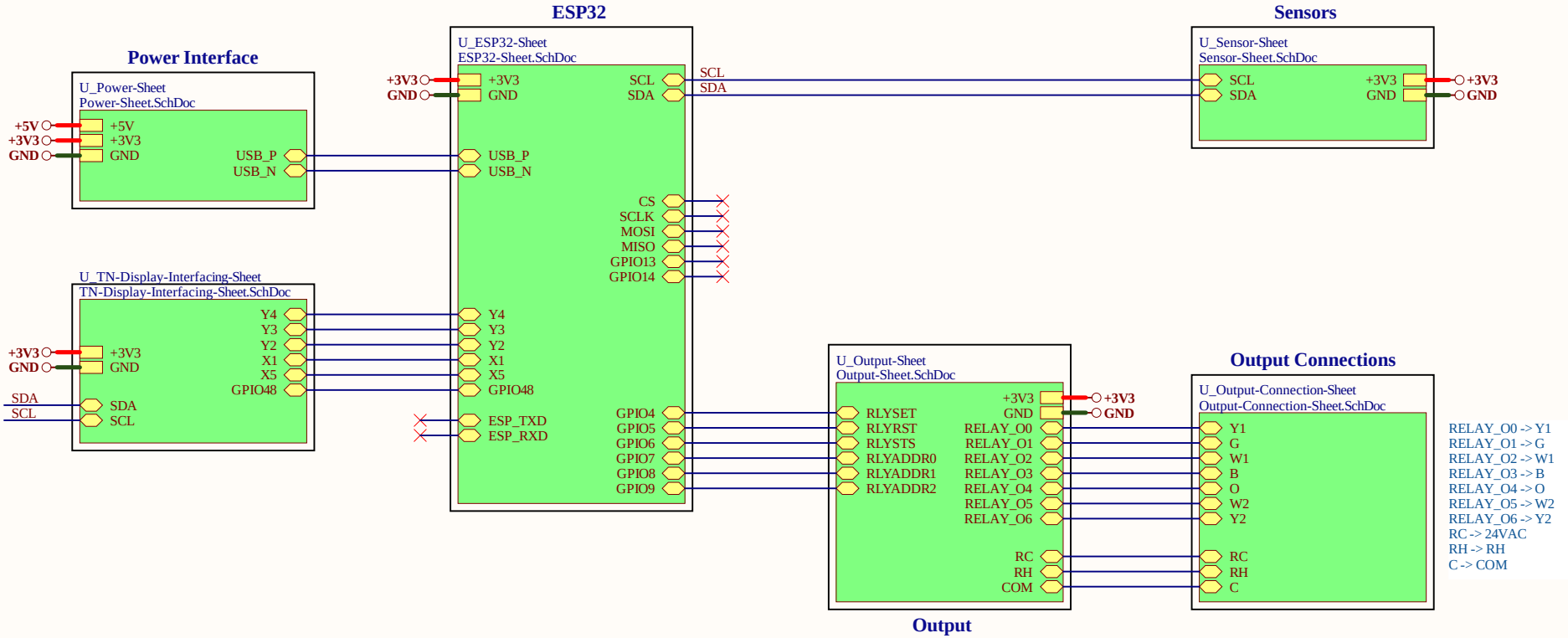


Relays, depending on the channel, switch between COM and 24VAC or RH wire.

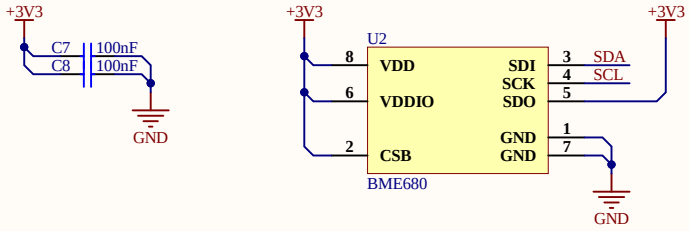
Y, G, O are switched betwen COM and 24VAC (RC)
W, B are switched between COM and RH wire

If C-wire is absent, 24VAC is absent, only W and B are switched to RH wire.

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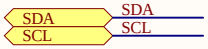
Title: Top-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 14

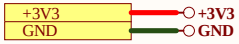
4-in-1 VOC Humidity Pressure Temperature



CSB -> VDDIO = I2C active
CSB -> GND = SPI active
I2C ADDR = BIN 1110111 HEX 0x77
I2C pull-up resistors are on the ESP32 sheet.

BSEC: Bosch Sensortec Environmental Cluster
BSEC precisely performs several calculations outside the device such as ambient air temperature, ambient relative humidity, pressure and air quality (IAQ) level.
https://www.bosch-sensortec.com/software-tools/software/bsec/
https://github.com/BoschSensortec/BSEC-Arduino-library

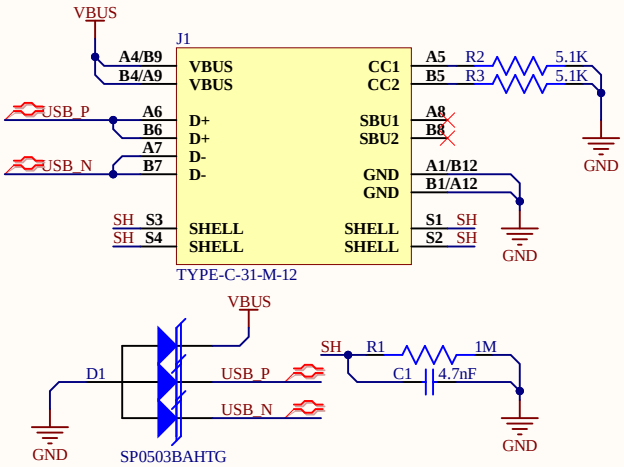




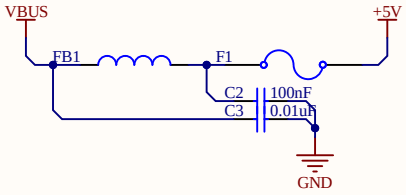
eSpark Consulting Group

Title: Sensor-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 14

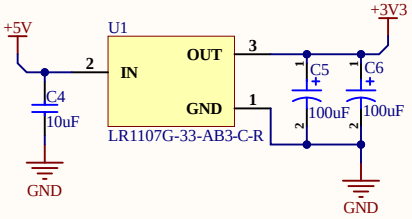
USB



Filters



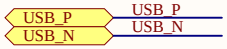
LDO 3.3V



Est. Consumptions:
Idle: 160 mA
Max: 1050mA (worst case!)

Test Points

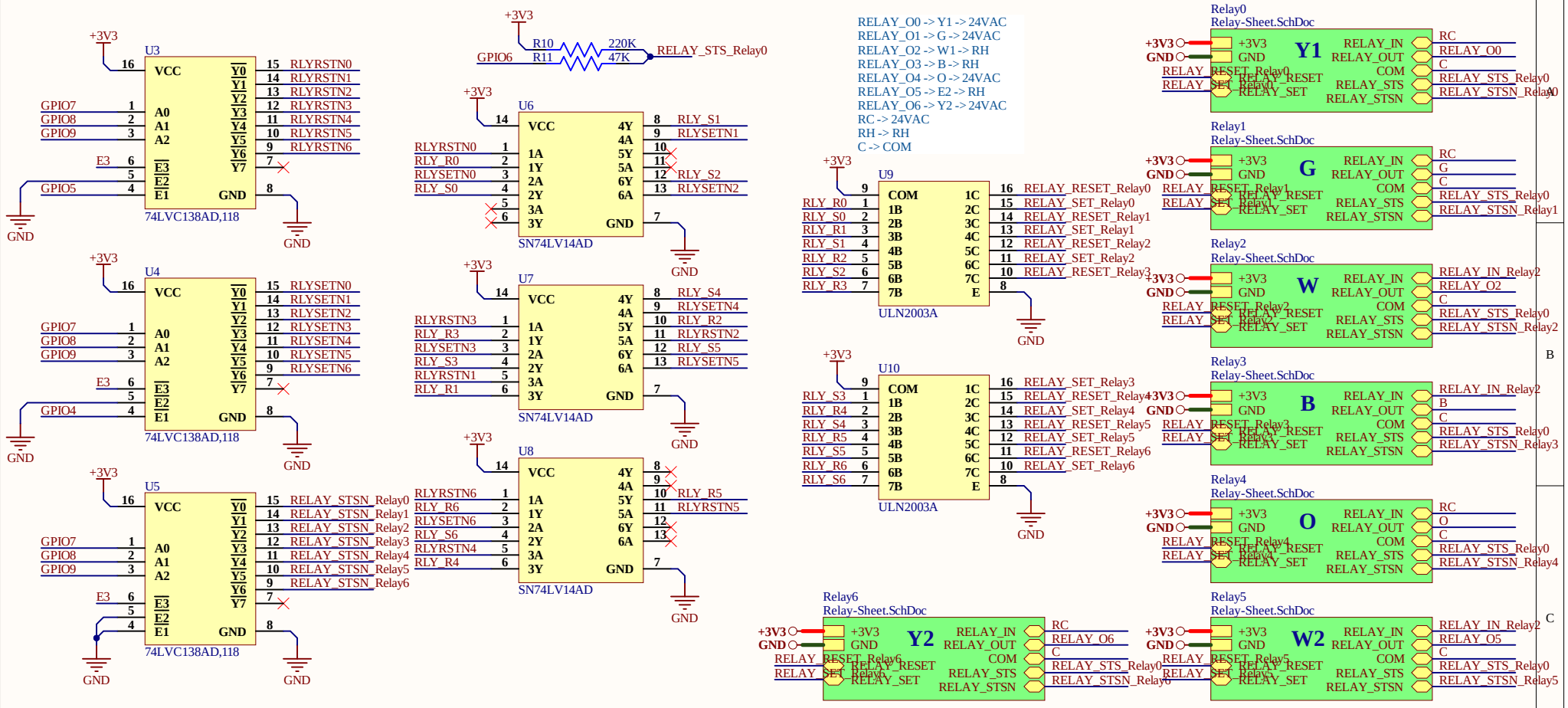


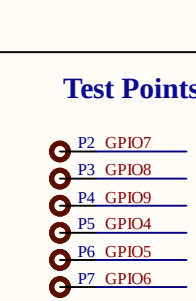


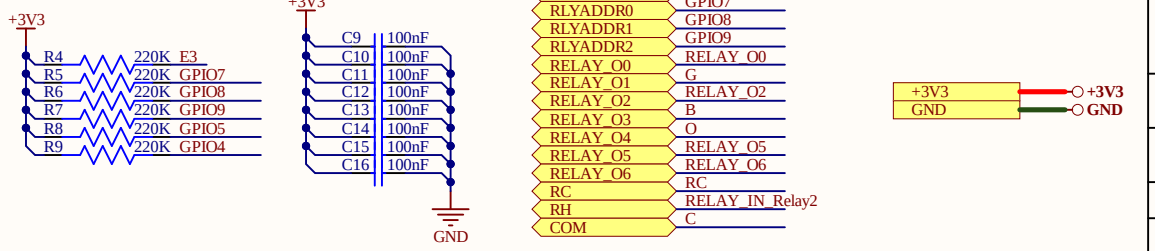


eSpark Consulting Group

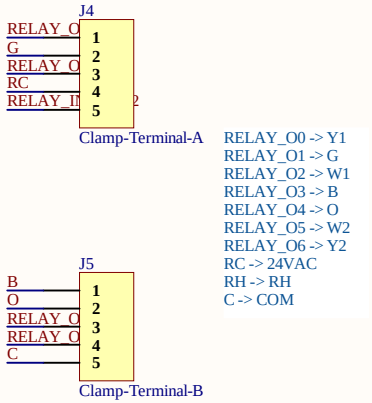
Title: Power-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 14

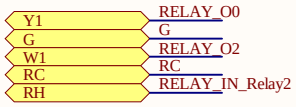


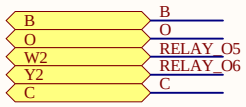




Title:	Output-Sheet.SchDoc
File:	ezlo-thermostat-basic.PrjPcb
Date: 25-Sep-23	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 14

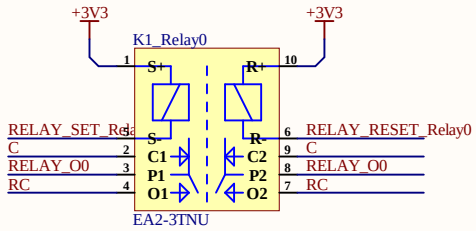


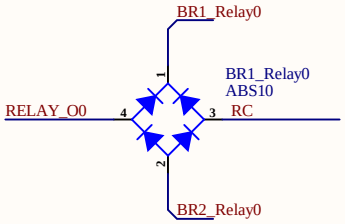


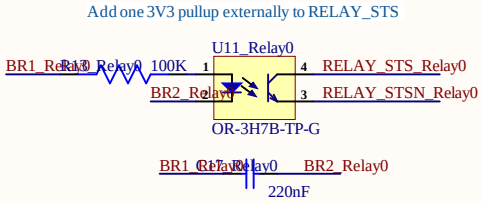


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Title: Output-Connection-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 14









Connect RELAY_IN to 24VAC or RH AC Input

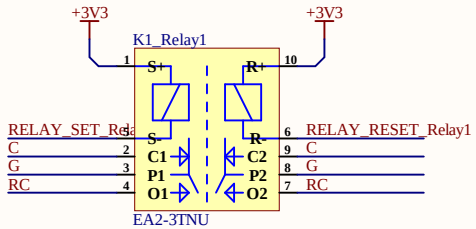


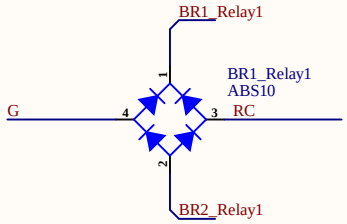


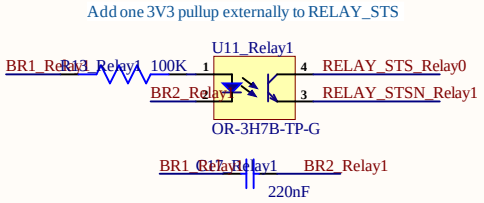


eSpark Consulting Group

Title: Relay-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 6.1
Drawn By: Huzaifa Ali	Page Total: 14



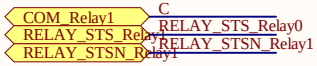






Connect RELAY_IN to 24VAC or RH AC Input

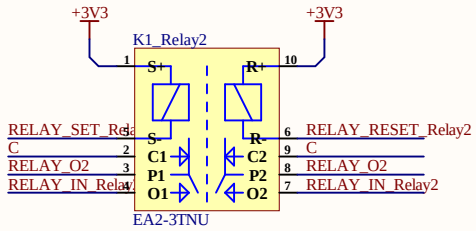


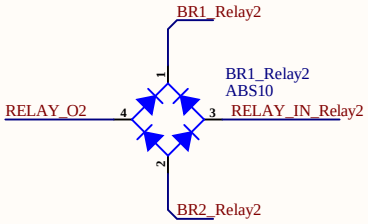




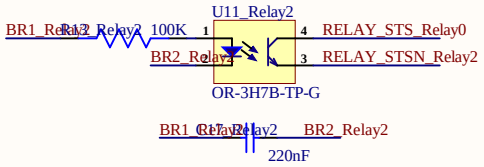
eSpark Consulting Group

Title: Relay-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 6.2
Drawn By: Huzaifa Ali	Page Total: 14





Add one 3V3 pullup externally to RELAY_STS





Connect RELAY_IN to 24VAC or RH AC Input

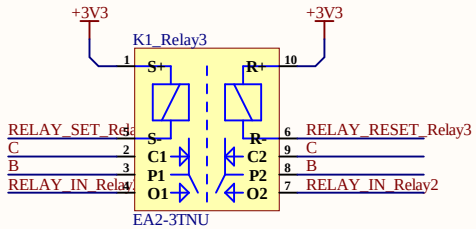


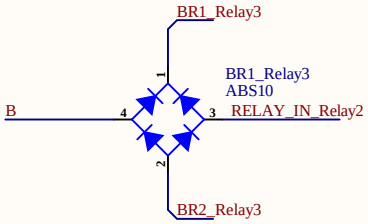




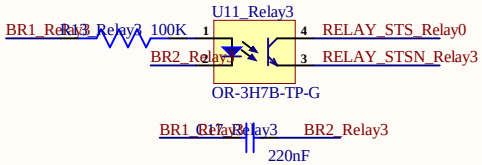
eSpark Consulting Group

Title: Relay-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 6.3
Drawn By: Huzaifa Ali	Page Total: 14





Add one 3V3 pullup externally to RELAY_STS





Connect RELAY_IN to 24VAC or RH AC Input

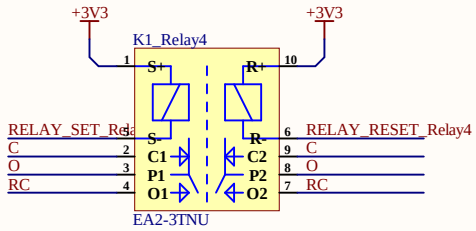


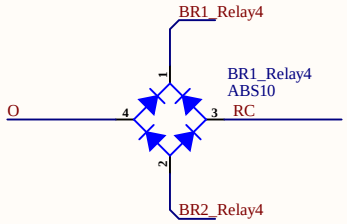


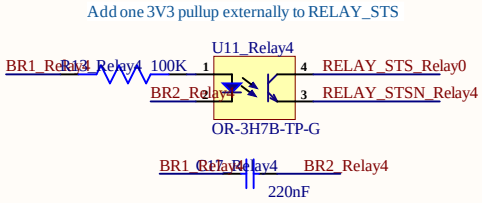


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Title: Relay-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 6.4
Drawn By: Huzaifa Ali	Page Total: 14



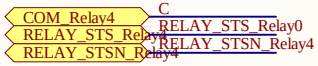






Connect RELAY_IN to 24VAC or RH AC Input





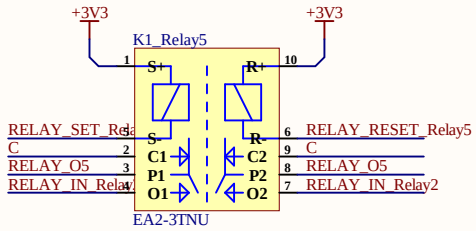


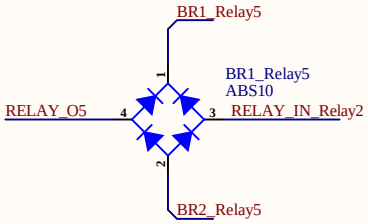
eSpark Consulting Group

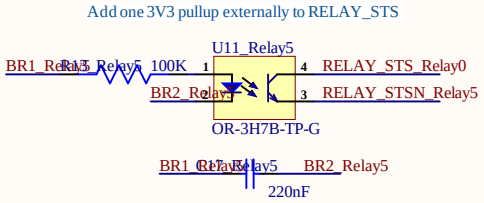
Title: Relay-Sheet.SchDoc

File: ezlo-thermostat-basic.PrjPcb

Date: 25-Sep-23	Page Num: 6.5
Drawn By: Huzaifa Ali	Page Total: 14









Connect RELAY_IN to 24VAC or RH AC Input

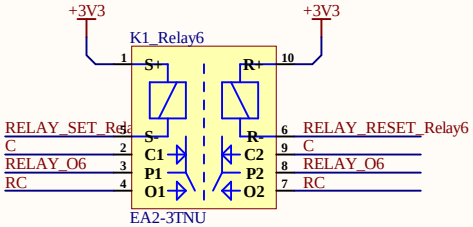


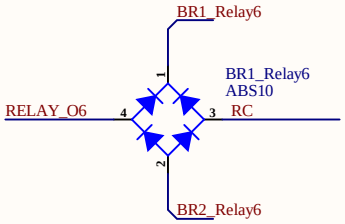


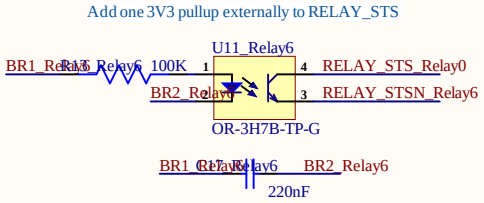


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Title: Relay-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 6.6
Drawn By: Huzaifa Ali	Page Total: 14









Connect RELAY_IN to 24VAC or RH AC Input



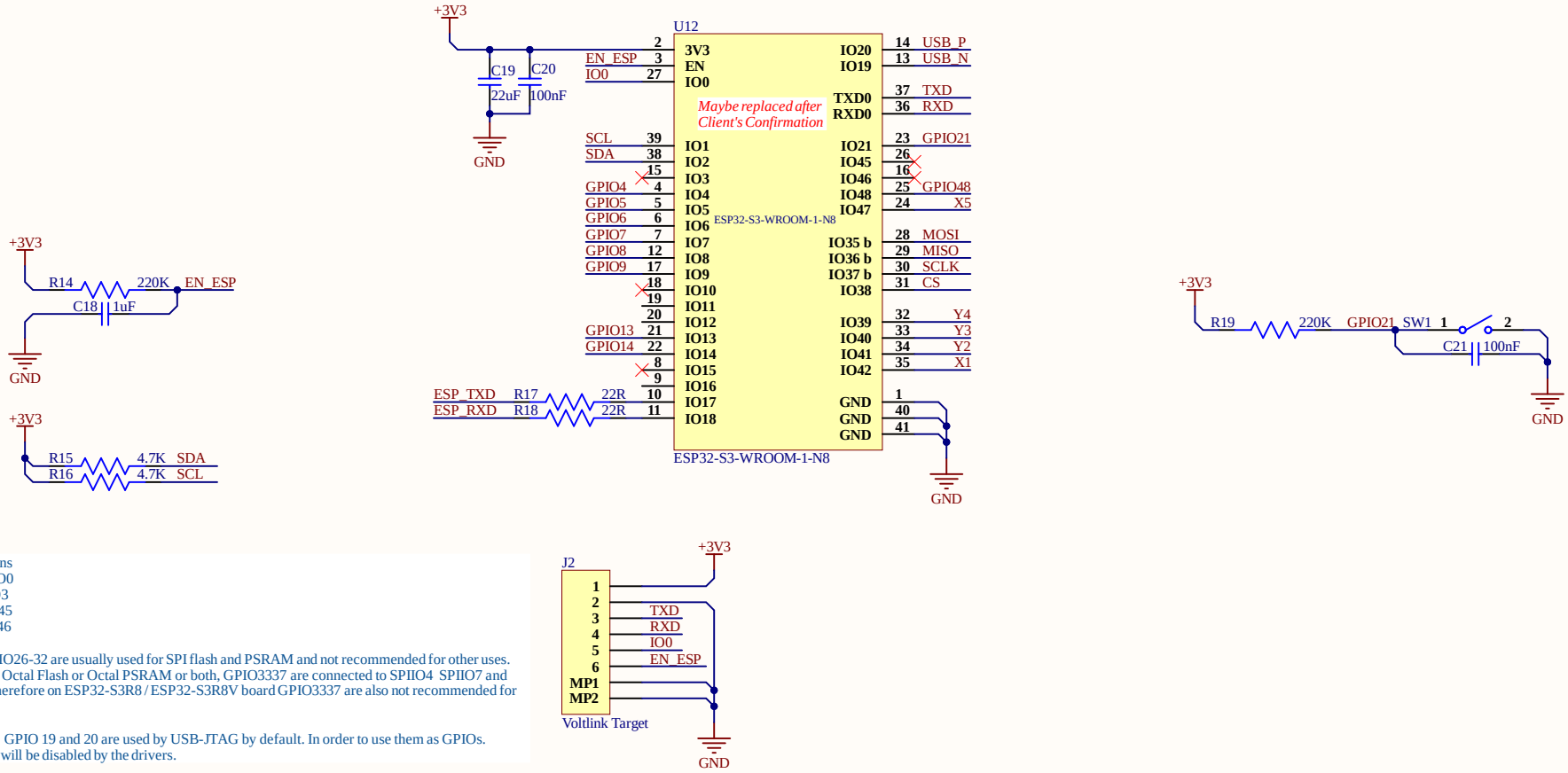




eSpark Consulting Group

Title: Relay-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 6.7
Drawn By: Huzaifa Ali	Page Total: 14

ESP32 MCU



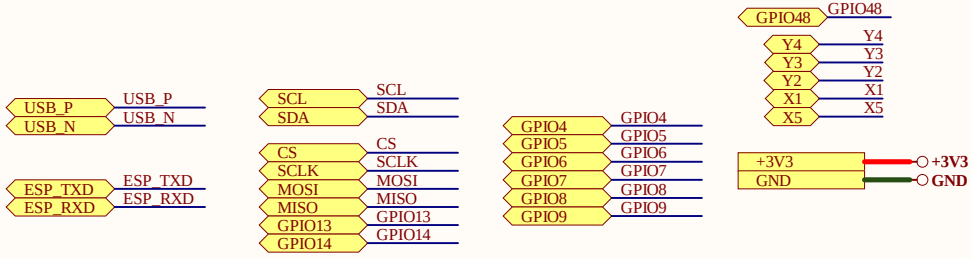
Strapping pins
- BOOT = IO0
- JTAG = IO3
- VSPI = IO45
- LOG = IO46

SPI0/1: GPIO26-32 are usually used for SPI flash and PSRAM and not recommended for other uses. When using Octal Flash or Octal PSRAM or both, GPIO3337 are connected to SPIIO4 SPIIO7 and SPIDQS. Therefore on ESP32-S3R8 / ESP32-S3R8V board GPIO3337 are also not recommended for other uses.

USB-JTAG: GPIO 19 and 20 are used by USB-JTAG by default. In order to use them as GPIOs. USB-JTAG will be disabled by the drivers.

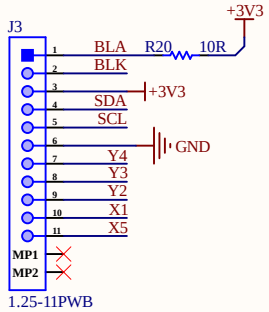
Test Points

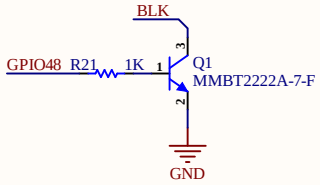
P8 SCL
P9 SDA
P10 MOSI
P11 MISO
P12 SCLK
P13 CS



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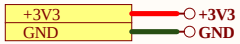
Title: ESP32-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 14











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Title: TN-Display-Interfacing-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 25-Sep-23	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 14